Fiscal Year:	FY 2018	**Task Last Updated:**	FY 04/03/2018
PI Name:	German, Randall M. Ph.D.		
Project Title:	Multi-Scale Modeling and Experimentation on Liquid Phase Sintering in Gravity and Microgravity Environments (80NSSC19K0231 ; NNX16AK21G)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	co-investigator Eugene A Olevsky, SDSU, Elisa Torresani, SDSU, MS 1326		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2010 Materials Science NNH10ZTT001N
Start Date:	07/01/2016	**End Date:**	06/30/2020
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Bhat, Biliyar	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 6/30/2020 with new grant number 80NSSC19K0231 per NSSC and MSFC personnel (Ed., 2/21/19)		
Key Personnel Changes/Previous PI:	none		
COI Name (Institution):	Olevsky, Eugene Ph.D. (San Diego State University)		
Grant/Contract No.:	80NSSC19K0231 ; NNX16AK21G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE (Ed., June 2016): Continuation of "Multi-Scale Modeling and Experimentation on Liquid Phase Sintering in Gravity and Microgravity Environments", grant #NNX14AB31G with the same Principal Investigator. NOTE (Ed., 2/21/19): Extended to 6/30/2020 with new grant number 80NSSC19K0231 per NSSC and Marshall Space Flight Center (MSFC) personnel. A research project is in progress to investigate the gravity-dependent densification, microstructure development, and the related macroscopic shape evolution (and distortion) of multi-phase powder materials during liquid phase sintering (LPS). The research is aimed at achieving two critical goals: (i) in-depth analysis of the LPS induced pore-grain structure evolution by the deconvolution of the gravity impact and (ii) exploring sintering under microgravity conditions as the basis for techniques applied to in-space fabrication and repair. Indeed, future NASA efforts to extend human exploration back to the Moon and beyond will require development of processes that permit fabrication under reduced gravity conditions. This capability is needed to reduce resource requirements for extraterrestrial construction and in space repair. Many proposed in-space fabrication and repair techniques involve particulate materials processing, such as via sintering, but limited data are available to assess the viability of these proposed net-shape techniques under reduced gravity. The research team has developed a computer code for the multiple scale simulation of sintering, embracing both microstructure development and macroscopic densification and associated shape distortion. Liquid phase sintering experiments on Earth and in microgravity will be conducted for a variety of tungsten heavy alloys (W-Ni-Cu-Mn) to validate the sintering model framework. The anticipated research outcomes will be relevant to current and future space exploration needs for habitat creation, extraterrestrial exploration, and vehicle repair activities during trips to Mars or the Moon.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research is focused on liquid phase sintering, one of the most widely used manufacturing routes for ceramics, cemented carbides, powder metals, and composites. In the USA this is responsible for approximately $25 billion in annual production of components used in metal cutting, oil well drilling, automotive engines, electric motor bearings, computer heat sinks, jet engine valves, and even wrist watches (especially the newest cell phone coupled devices). These components are designed for high performance and high precision, and are often fabricated in large production quantities. The net-shape fabrication, where final size and shape is achieved without machining, is a challenge since often dimensions change 15 to 20% due to shrinkage in going from the powder to the dense component. During that high temperature sintering the component strength drops considerably and is subject to gravity effects. Thus, computer models of the sintering process are vital to proper tool and process design. Further, to attain specific mechanical, thermal, electrical, or other properties requires further attention to microstructure control. This research provides a verified computational model to enable prediction of final size and shape for different gravitational fields, using a multiple scale model with hopes to extend from geometry to property predictions. The robust inclusion of gravitational gradients, liquid-solid migration, and substrate friction, the computer simulation will be useful for predicting component attributes over a range of gravitational conditions, as required for eventual additive manufacturing as potentially envisioned for fabrication on the moon, other planets, or in microgravity, for example, in building structure on the lunar surface or repairing space vehicles. Of course, this has benefit on Earth with the evolution of a computational means to predict high tolerance shapes by sintering. We are adding the missing gravitational term to computer simulations, typically finite element analysis tools, for predicting final shape and size.
	The flight experiments are in progress. Samples were launched 2 April 2018 on SpaceX CRS-14. Up to now we have been limited to Earth-based sintering (1g). The results from ground experiments clearly show densification prior to distortion. The pending sintering in microgravity will help fill in time-based data (using short hold times and long hold times) to understand why incomplete densification arises. In the absence of pore buoyancy and grain compression, there is more distortion in microgravity. Our studies will examine the missing stress terns on particle-particle contacts. If habitats are going to be constructed by advanced robotic factories on the Moon or Mars, then problems will occur since up to now we are unable to anticipate density, properties, or defects when sintering in space or anything other than normal gravity. These points justified the current experimental program. We are focused on a few questions: How does gravity influence sintering densification and distortion? Can this behavior be predicted? How will this be included in computer simulations? Will it be possible to conduct meaningful extraterrestrial sintering? The launch of samples for processing on the International Space Station (ISS) is a major advance in 2018. Since the start in 2010, ground experiments and laboratory work led to flight qualification mode in late 2012. Subsequently, interface with the LGF (low gradient furnace) required design of SCA (sample cartridge assembly) tubes containing the LPS (liquid phase sintering) samples. Several issues were addressed and difficulties overcome to allow flight hardware launch in April 2018. The European Space Agency (ESA)-provided LGF is a facility consisting of a core furnace with vacuum, power, SCA or sample insertion, and units for water and process gas. Cartridges containing the samples are packaged and simply inserted in the furnace for each run. The LGF has both temperature stability and uniformity, nominally ± 0.5°C. The bore is 30 mm with 250 mm heated length; the 30 mm diameter limits sample size, the flight samples are 12 mm diameter. The furnace maximum temperature of 1200°C as set by thermocouple life. This limitation mandated the search for a tungsten alloy that densifies at 1200°C. The development of a W-Ni-Cu-Mn heavy alloy for low temperature liquid phase sintering was a significant effort and is detailed in our early publications. This alloy is compatible with the furnace temperature capability. The density difference between solid (19.2 g/cc), liquid (10 g/cc), and porosity (0 g/cc) accentuates the role of gravity in small samples. Commercial tungsten, nickel, copper, and manganese powders were used. The matrix ingredients that form the liquid during sintering, were mixed in the desired ratios, mechanically alloyed, and then milled intensively with tungsten to homogenize each composition. Logs were formed using cold isostatic pressing at 203 MPa. The logs were presintered for 60 min (with heating hold of 60 min at 900°C) at 1050°C in hydrogen, giving 63% density. The logs were then machined into 12 mm diameter by 12 mm high right circular cylinders with typically 13 g mass at an average 9.8 g/cm3 density. Each sample was cleaned (ethanol), dried, measured for height, diameter at four 45° rotation locations, mass, and vacuum desiccated. Laboratory sintering trials clearly show solid migration to the bottom, liquid migration to the top, and considerable pore

Task Progress:	annihilation. In microgravity we expect difficulty with densification due to pore coalescence and the absence of pore buoyancy forces. Also, in µg the solid skeleton has less structural rigidity since the solid grains have reduced interparticle compression. The nominal sintering force is 2.2 MPa acting to pull the particles together. Along the vertical axis the 12 mm sample height of nominally 9.8 g/cc alloy contributes stress at the particle contacts due to gravity in the form of an interparticle compression stress that is inverse with the particle size near 5 µm. The gravity induced particle bonding compression stress from gravity is a significant additional stress acting on the particle contacts oriented perpendicular to the gravitation vector. As the interparticle bonds grow, the enlargement of bonds between the particles reduces the compressive stress. Significantly more sintering and early particle bonding is expected in the presence of gravity, leading to more structural rigidity to resist distortion while also contributing pore buoyancy to eliminate porosity. When gravity is missing, there still is a sintering stress due to capillarity. But the loss of compressive stress on the structure takes away the strong solid-solid bonds that resist distortion. The degree of distortion should vary with the dihedral angle. The updated design of experiments relies on the tungsten content to set the solid:liquid ratio at the sintering temperature (roughly the wt. % matrix times 2 gives the vol. % liquid; 30 wt. % results in 50 to 56 vol. % matrix). The alloys range from 70 to 90 wt. % W. The Ni:Cu ratio is adjusted to 1.5, 1, or 0.67 to change the dihedral angle. Six hold times are employed to capture the sintering evolution. In dilatometry trials and laboratory batch sintering trials, up to 1200°C took the high liquid content (70 and 80 wt. % W) alloys to essentially full density in 60 min of vacuum sintering at 1200°C. Short microgravity experiments are not useful in isolating the gravity effect since the viscosity of the tungsten alloys varies during densification, dropping to about that of a thick polymer. The flight test matrix consists of 70, 80, 85, and 90 wt. % tungsten samples. The matrix involves then one of three compositions Ni:Cu:Mn at 1.2:0.8:1 (L for low Cu and low dihedral angle), 1:1:1 (B or base composition), or 0.8:1.2:1 (H for high Cu and high dihedral angle). The Mn lowers the sintering temperature while the Ni:Cu ratio adjusts the dihedral angle. Tungsten is more soluble in nickel-rich liquids, so the high nickel content will allow more grain boundary wetting (low dihedral angle) and the low nickel content then gives a high dihedral angle. Cartridge C0 is designed to reach the temperature (about 1180°C), where dilatometry first indicates densification. Cartridges C1 to C6 provide time increments near 1200°C to nominally capture the sintering trajectory up to an hour of sintering. Actual times are given later based on finite elemental thermal analysis pegged to ground sintering trials. Trials were conducted to prove out the hardware, cartridge integrity, and experiment design. Early problems included samples sticking to the crucible, largely cured by more care in sample placement and cartridge handling. Some early quartz containers ruptured, and this was cured by adding fibrous packing in at the end of the sample stack. Instrumentation provided data for thermal analysis, and that identified a role of the adiabatic zone temperature on the temperature profile in the crucible stack. Off-gassing was a concern, as cartridge gas emission peaked just as the sintering temperature was reached. Since much of the cartridge is not involved in sample containment, the decision was to lower the adiabatic zone temperature to 1050°C to reduce gas loads. Finally, oxidation of the manganese is a concern, where the oxygen is transferred from the tungsten to manganese, since this is a volatile oxide. Samples from those trials were analyzed for density, distortion, and microstructure. The real payoff comes when microgravity sintered samples are returned for direct comparison. After passing these tests, the experiments moved to flight status. Distortion and densification are documented for the sintered sample. Dimensions are captured using a touch probe coordinate measuring machine. The selection of samples for flight versus ground was largely random. Now with the flight experiments underway, the research team is focused again on the computer simulations and updates related to inclusion of the gravity effects for ground sintering. Samples returned from microgravity sintering will provide data on the sintering trajectory and enable modifications in the constitutive equations are appropriate for microgravity. These are exciting times now that flight experiments are starting. The following papers have been submitted for publication: Alvarado-Contreras, J, Maniere, C. German, RM, Olevsky, EA "Multi-scale modeling of various modes of densification of hierarchical powder structures" Submitted Acta Materialia, 2018, submitted and pending review and publication, as of April 2018. Torresani E, German RM, Olevsky EA. "Gravity effects during sintering of additively manufactured powder components." Acta Materialia, 2018, submitted for publication as of April 2018.
Bibliography Type:	Description: (Last Updated: 03/27/2025)
Papers from Meeting Proceedings	German RM, Olevsky EA. "How gravity influences accuracy in liquid phase sintering." Presented at the 19th Plansee Seminar - International Conference on Refractory Metals and Hard Materials, Reutte, Austria, May 29- June 1, 2017. In: Plansee Proceedings, L. S. Sigl, H. Kestler, and A. Pilz (eds.), Reutte, Austria, 2017. p. RM30.01-30.05. , Jun-2017
Papers from Meeting Proceedings	German RM, Olevsky EA. "Gravitational Role in Dimensional Control in Liquid Phase Sintering." International Conference on Sintering 2017, San Diego, California, November 12-16, 2017. Proceedings, International Conference on Sintering 2017. Published by American Ceramic Society, Westerville, Ohio. , Nov-2017
Papers from Meeting Proceedings	Olevsky EA. "Challenges and further developments in modeling of sintering." RM German Honorary Symposium, International Conference on Sintering 2017, American Ceramic Society, San Diego, CA, 2017. RM German Honorary Symposium, International Conference on Sintering 2017, San Diego, California, November 12-16, 2017. , Nov-2017